Domain-Adversarial Training Under Label Shift Can Silently Invert Out-of-Distribution Detection

Tran Duc Vinh VINHTRAN9969@GMAIL.COM
Hanoi University of Science and Technology [PENDING: verify author list and affiliations]

Editors: Under Review for MIDL 2027

Abstract

Domain-adversarial training (DANN) is a standard remedy for site and device shortcuts in medical imaging, and deployed models increasingly rely on out-of-distribution (OOD) monitors to flag unfamiliar inputs. We show that the two can conflict silently. On skin-lesion classification with a lesion-disjoint ISIC 2019 split, gradient reversal toward smartphone images drives Mahalanobis OOD detection from AUROC 0.825 to 0.381 (95% CI $[0.358, 0.404]$), and below chance on three clinical datasets no part of training saw (Fitzpatrick17k 0.405; MED-NODE and SD-198 [PENDING: last-checkpoint values]), while in-distribution balanced accuracy stays at 0.47–0.48 [PENDING: paired CI, calibration]. The inversion is caused by label shift between the domains the adversary aligns: changing only the class mix the adversary sees moves unseen-domain AUROC by 0.24–0.37, and the effect persists at matched domain leakage. Mechanistically, DANN pulls unfamiliar images toward the majority class; under label shift it reaches chance-level unweighted domain leakage by misaligning classes, while the domain stays decodable within each class. The consequence is clinical: the share of melanomas from unseen sources predicted benign nevus with confidence above 0.9 rises from 8% to 49–81%. Importance-weighted alignment helps in proportion to how well the target class priors are known: with label-free BBSE estimates it recovers 32–45% of the lost AUROC at matched leakage, and with priors from 50 labelled target images detection returns to its pre-adaptation level, at a cost of 0.08 in-distribution balanced accuracy. OOD monitoring should be re-validated after domain adaptation, and domain leakage should be measured within class.

Keywords: domain-adversarial training; out-of-distribution detection; label shift; calibration; skin lesion classification; deployment monitoring

1. Introduction

Skin lesion classifiers learn acquisition artefacts as readily as pathology: ink markings, rulers, dermatoscope type and contributing site all correlate with diagnosis in public datasets (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often by domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). A deployed model also needs a second safeguard: an out-of-distribution (OOD) monitor that flags inputs unlike its training data so that a clinician reviews them. This paper asks what invariance training does to that monitor.

In our setting it can turn the monitor upside down. As the adversarial weight grows, OOD detection does not merely weaken; it falls below chance, so that images from clinics the model has never seen score as *more* familiar than its own test data. Nothing in a standard evaluation reveals this: in-distribution accuracy and calibration stay flat, and the

© 2027 CC-BY 4.0, T.D. Vinh.

domain-leakage probe, the quantity the method is meant to reduce, improves. The cost falls on the patients a monitor exists to protect: under the same training, melanomas from unseen clinics are predicted benign nevus with confidence above 0.9 up to ten times more often, and the OOD monitor that should route them to a clinician ranks them as routine.

We trace the failure to label shift. ISIC 2019 is 50.8% nevus, the smartphone images the adversary aligns against are 12.3% nevus, and matching the marginal feature distributions of domains with different class priors forces the representation to misplace one domain's classes (Zhao et al., 2019; Tachet des Combes et al., 2020). We show the consequence for OOD monitoring with an intervention that changes only the class mix the adversary sees, and we locate the mechanism in the feature space.

Contributions.

1. **The failure.** Under a lesion-disjoint split, gradient reversal inverts Mahalanobis OOD detection on held-out smartphone images and on three clinical datasets that no branch of training saw, with flat in-distribution accuracy and calibration, under two training recipes whose accuracy is within 0.06 of a standard ERM baseline on the same split.

2. **The cause.** Setting the class mix seen by the adversary to that of the target removes the inversion on unseen domains (AUROC difference 0.24–0.37); a regression at matched domain leakage attributes the effect to label shift rather than to deeper invariance.

3. **The mechanism and a measurement lesson.** DANN pulls images that differ from the source along the domain direction toward the majority class. Under label shift, an unweighted domain probe reports chance-level leakage while a within-class probe still reads the domain: invariance should be measured within class.

4. **Consequence and mitigation.** Melanomas from unseen sources are called benign nevi with high confidence up to ten times more often. Importance weighting with label-free prior estimates recovers part of the loss; with priors from 50 labelled target images it recovers fully, at a cost in in-distribution accuracy.

2. Related work

Shortcuts and invariance in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion masked out (Bissoto et al., 2020), and performance degrades on darker skin (Daneshjou et al., 2022). Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) are the usual remedies; our dual-encoder model is a domain separation network and is a vehicle for the intervention, not a contribution.

Invariance under label shift. When class priors differ between domains, matching marginal feature distributions cannot also keep the joint error small (Zhao et al., 2019); importance-weighted alignment (IWDAN, IWCDAN) corrects for this with estimated target class priors (Tachet des Combes et al., 2020), which can be estimated without target labels by black-box shift estimation (Lipton et al., 2018) or EM (Saerens et al., 2002). These

results concern classification error. We study what the same misalignment does to OOD monitoring, and show that a standard domain-leakage probe misreports invariance under label shift.

OOD detection after adaptation. Full-spectrum OOD detection (Yang et al., 2023; Zhang et al., 2023) asks that covariate-shifted in-distribution inputs be accepted; our smartphone and clinical images fall in that cell, and Section 5 argues that acceptance is safe only when the model also classifies them. Calibration degrades under shift (Ovadia et al., 2019) and after domain adaptation (Wang et al., 2020). Closest to our observation, Wu et al. report Mahalanobis and ODIN AUROC below 0.5 on rotated MNIST after adaptation with an MMD-based method (DAN) (Wu et al., 2023), as a baseline in a study of continually adaptive detection. We show the inversion on medical images, identify label shift as its cause, and characterise the mechanism.

3. Materials and methods

3.1. Data and splits

ISIC 2019 (Tschandl et al., 2018) (25,331 dermoscopic images, eight classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC) is the in-distribution source. We split it 60/20/20 into train, validation and test *grouped by* `lesion_id` (16,211 / 4,053 / 5,067 images), because 60% of test images in the image-level split share a lesion with training (Tran, 2027). **PAD-UFES-20** (Pacheco et al., 2020) (2,298 smartphone photographs, six of the eight classes) is the domain the adversary aligns against. Its labels are never used in any loss. We split it by patient into `pad_adv` (1,582 images, 961 patients), seen by the adversarial and context branches, and `pad_heldout` (716 images, 412 patients), never seen.

Unseen domains. Four clinical-photograph datasets enter no part of training: Fitzpatrick17k (Groh et al., 2021) (3,887 images obtained), MED-NODE (170 images), SD-198 (6,584 images) and SCIN (10,406 images, 5,033 cases) [PENDING: cite MED-NODE, SD-198, SCIN; licences: SCIN CC BY 4.0, MED-NODE none stated, SD-198 official source]. SCIN is mostly inflammatory conditions; only 0.6% of its images map to the ISIC classes, so we use it only as an OOD set for detection and report it separately; no label-based analysis is done on SCIN. Images are cropped to a lesion-centred box and resized to 224×224.

3.2. Model and training

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, z_ℓ^{norm}) used for diagnosis through a linear head, and a context latent $z_c \in \mathbb{R}^{64}$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the latents. AdamW (learning rate 10^{-4}; adversary $\times 30$, classifier $\times 0.2$), 40 epochs, domain-balanced batches. We sweep $\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ with everything else fixed (seeds 42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 elsewhere). A second recipe (A4: $z_\ell \in \mathbb{R}^{64}$, classifier $\times 1$, class-weighted loss) serves as a robustness check.

Checkpoints. We report the last checkpoint. Validation balanced accuracy (eight classes, chance 0.125) has a lesion-clustered bootstrap SE of 0.023–0.026, and the best-validation and last checkpoints differ by less than one SE, so best-validation selection picks early epochs by noise. Best-validation values are given for comparison.

Competence. In-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ is 0.496 (main recipe) and 0.520 (A4), 0.04–0.06 below a standard single-encoder ERM on the same split (0.556 [PENDING: ERM reference at 5 seeds, with SE]) and four times chance (0.125). Lesion-disjoint eight-class ISIC 2019 is hard; the image-level split inflates the same model's balanced accuracy by 0.20 (Tran, 2027).

3.3. Interventions on label shift

Class mix seen by the adversary. Keeping the classification loss on all of ISIC, we resample only the ISIC images fed to the adversarial branch so that their class mix is ISIC's own (100%, the default), halfway to PAD's (50%), or PAD's (0%, no label shift between the aligned domains), at $\lambda_{\text{adv}} \in \{0.5, 2\}$ with five seeds (three at 50%).

Importance-weighted alignment. IWDAN and IWCDAN (Tachet des Combes et al., 2020) reweight the source in the adversarial loss by estimated target class priors; we compare BBSE estimates (Lipton et al., 2018), EM with temperature-scaled probabilities (Saerens et al., 2002), priors from 50 labelled `pad_adv` images (a few-label setting), and the true `pad_adv` priors (oracle, not deployable). CDAN (Long et al., 2018) is a class-conditional adversarial baseline.

3.4. Measurements

OOD detection. Class-conditional Mahalanobis with shared covariance (Lee et al., 2018) (primary), k-NN (Sun et al., 2022), maximum cosine to a class centroid, maximum softmax probability (Hendrycks and Gimpel, 2017) and energy (Liu et al., 2020), all fit on ISIC train. AUROC treats OOD as the positive class, so 0.5 is chance and values below 0.5 mean OOD images score as more familiar than ID test images. The five scores are functions of the same latent, so they are not independent evidence.

Inversion. We call a cell inverted when the cluster-bootstrap 95% CI of the seed-mean Mahalanobis AUROC lies below 0.5 and at least three seeds' own intervals do (all three in cells with three seeds). Bootstrap draws (2,000) resample ISIC test by lesion and OOD sets by patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k) or case (SCIN); MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow. Runs with in-distribution balanced accuracy below 0.25 are excluded as degraded and reported.

Domain leakage. Balanced accuracy (chance 0.5) of a logistic probe for the source dataset on frozen z_ℓ^{norm}. The probe used in earlier work scores ISIC test against all of PAD, including images the adversary trained on. We therefore also report probes on held-out images only (linear, one-hidden-layer MLP, k-NN), a class-balanced probe that reweights both domains to a common class mix, and a within-class probe averaged over classes.

Table 1: The adversarial-weight sweep (main recipe, lesion-disjoint ISIC split, last checkpoint). Mean ± s.d. over five seeds per λ_{adv} (42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 otherwise). Leakage: linear domain probe on held-out images (chance 0.5). ID: balanced accuracy, eight classes (chance 0.125). Mahalanobis AUROC, OOD positive; k: seeds whose interval lies below 0.5. Best-val: the best-validation checkpoint, for comparison.

λ_{adv}	leakage	ID bal. acc	`pad_heldout`	k	Fitzpatrick17k	k	`pad_heldout` (best-val)
0	0.842 ± 0.050	0.481 ± 0.016	0.825 ± 0.022	0/5	0.712 ± 0.013	0/5	0.808 ± 0.027
0.25	0.502 ± 0.002	0.481 ± 0.010	$\mathbf{0.445 \pm 0.022}$	4/5	$\mathbf{0.396 \pm 0.032}$	5/5	0.525 ± 0.121
0.5	0.508 ± 0.010	0.484 ± 0.005	$\mathbf{0.415 \pm 0.016}$	5/5	$\mathbf{0.377 \pm 0.046}$	5/5	0.426 ± 0.008
1	0.505 ± 0.005	0.482 ± 0.017	$\mathbf{0.390 \pm 0.022}$	5/5	$\mathbf{0.388 \pm 0.039}$	5/5	0.387 ± 0.016
2	0.507 ± 0.009	0.472 ± 0.018	$\mathbf{0.381 \pm 0.023}$	5/5	$\mathbf{0.405 \pm 0.031}$	5/5	0.403 ± 0.028
4	0.527 ± 0.035	0.480 ± 0.016	$\mathbf{0.416 \pm 0.034}$	5/5	$\mathbf{0.354 \pm 0.066}$	5/5	0.433 ± 0.020
8	0.520 ± 0.028	0.480 ± 0.018	$\mathbf{0.439 \pm 0.020}$	5/5	$\mathbf{0.383 \pm 0.036}$	5/5	0.454 ± 0.044

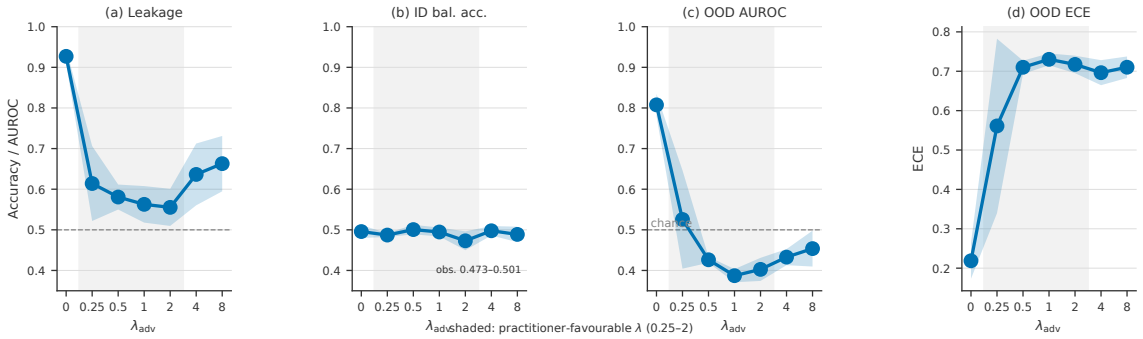


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

Calibration and diagnosis. Expected calibration error (15 bins) on ISIC test and on `pad_heldout` (six shared classes), and zero-shot balanced accuracy on `pad_heldout`. Out-of-domain ECE is close to mean confidence minus accuracy here, so these are views of one phenomenon.

4. Results

4.1. OOD detection inverts while in-distribution metrics stay flat

The sweep. At the last checkpoint, Mahalanobis AUROC on `pad_heldout` falls from 0.825 at $\lambda_{\text{adv}} = 0$ to 0.445 at $\lambda_{\text{adv}} = 0.25$ and stays between 0.381 and 0.445 at every larger weight, meeting the inversion criterion at all six non-zero weights; on Fitzpatrick17k it falls

Table 2: Unseen domains: Mahalanobis AUROC at $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$, mean ± s.d. over five seeds, with the number of seeds whose interval lies below 0.5. Main recipe, best-validation checkpoint [PENDING: MED-NODE/SD-198/SCIN at the last checkpoint]; A4, last checkpoint. MED-NODE and SD-198 are resampled by image. –: not bootstrapped.

recipe	set	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$	seeds < 0.5	inverted
main	`pad_heldout`	0.808 ± 0.027	0.403 ± 0.028	5/5	yes
main	Fitzpatrick17k	0.717 ± 0.034	0.402 ± 0.018	5/5	yes
main	MED-NODE	0.536 ± 0.037	0.306 ± 0.071	5/5	yes
main	SD-198	0.713 ± 0.037	0.375 ± 0.026	5/5	yes [PENDING: SD-198 cluster CI]
main	SCIN	0.726 ± 0.021	0.559 ± 0.034	0/5	no
A4	`pad_heldout`	0.786 ± 0.019	0.452 ± 0.014	[PENDING: k]	yes
A4	Fitzpatrick17k	0.821 ± 0.023	0.458 ± 0.061	[PENDING: k]	yes

from 0.712 to 0.354–0.405 (Table 1). At $\lambda_{\text{adv}} = 2$ the seed-mean interval is $[0.358, 0.404]$ on `pad_heldout` and $[0.392, 0.419]$ on Fitzpatrick17k. In-distribution balanced accuracy stays within 0.472–0.484 [PENDING: paired $\lambda_{\text{adv}} = 2 - \lambda_{\text{adv}} = 0$ CI and ID/OOD ECE at the last checkpoint]. All five scores fall below 0.5 on `pad_heldout` at $\lambda_{\text{adv}} = 2$ (k-NN 0.427, cosine 0.420, MSP 0.455, energy 0.468; best-validation checkpoint); the standard k-th-neighbour k-NN gives the same values within 0.01. The best-validation checkpoint gives the same picture from $\lambda_{\text{adv}} = 0.5$ on, but at $\lambda_{\text{adv}} = 0.25$ it selects an earlier epoch and the inversion is not yet visible (0.525 ± 0.121).

Unseen domains. The inversion is not confined to the domain the adversary was trained against (Table 2). On three clinical datasets that entered no part of training, Mahalanobis AUROC falls from 0.717 to 0.402 (Fitzpatrick17k), 0.536 to 0.306 (MED-NODE) and 0.713 to 0.375 (SD-198) at $\lambda_{\text{adv}} = 2$, each meeting the inversion criterion. On SCIN, whose images are mostly inflammatory conditions outside the ISIC label space, detection falls from 0.726 to 0.559 without inverting.

A second recipe. With a 64-dimensional latent, unweighted cross-entropy replaced by a class-weighted loss and the last checkpoint (A4), detection again inverts at $\lambda_{\text{adv}} = 0.5$ and $\lambda_{\text{adv}} = 2$ on `pad_heldout` (0.468 $[0.440, 0.495]$; 0.452 $[0.425, 0.478]$) and Fitzpatrick17k (0.444; 0.458), with in-distribution balanced accuracy unchanged (0.520, 0.520, 0.510). On `pad_heldout` all five scores' intervals lie below 0.5; on Fitzpatrick17k k-NN (0.521) and energy (0.516) only collapse. A plain single-encoder DANN collapses to chance (0.51–0.56) without inverting (Appendix A.3).

4.2. Label shift between the aligned domains causes the inversion

Intervention. Changing only the class mix of the ISIC images the adversary sees switches the inversion on unseen domains on and off (Table 3). At $\lambda_{\text{adv}} = 2$, with the adversary's ISIC images resampled to PAD's class mix (shift 0%), Mahalanobis AUROC is 0.691 on Fitzpatrick17k, 0.541 on MED-NODE and 0.694 on SD-198, none inverted; with ISIC's own

Table 3: Label-shift intervention: Mahalanobis AUROC (main recipe, last checkpoint) by the class mix the adversary sees. Shift 100%: ISIC's own mix; 0%: PAD's mix. Leakage: probe of Table 1. Shift 0% at $\lambda_{\text{adv}} = 2$: four seeds after excluding one degraded run (in-distribution balanced accuracy $0.139 < 0.25$); shift 50%: three seeds.

λ_{adv}	shift	leakage	`pad_heldout`	Fitzpatrick17k	MED-NODE	SD-198
0	–	0.928	0.825	0.712	[PENDING:]	[PENDING:]
0.5	0%	0.790	0.499	0.674	[PENDING:]	[PENDING:]
0.5	50%	0.629	0.490	0.482	[PENDING:]	[PENDING:]
0.5	100%	0.628	0.361	0.385	[PENDING:]	[PENDING:]
2	0%	0.791	0.458	0.691	0.541	0.694
2	50%	0.658	0.454	0.441	[PENDING:]	[PENDING:]
2	100%	0.634	0.368	0.340	0.298	0.325

mix (shift 100%) it is 0.340, 0.298 and 0.325, inverted in every seed. Paired by seed, removing the label shift raises AUROC by 0.361, 0.243 and 0.372. The intermediate level falls between the two (Fitzpatrick17k 0.482 at $\lambda_{\text{adv}} = 0.5$, 0.441 at $\lambda_{\text{adv}} = 2$). On `pad_heldout`, the adversary's own target domain, a mild inversion remains without label shift (0.458; paired difference +0.093). One of five shift-0% runs collapsed (in-distribution balanced accuracy 0.139) and is excluded.

Matched leakage. Removing label shift also leaves more domain information in the latent (leakage 0.791 vs 0.634), so the intervention alone does not separate label shift from depth of invariance. A regression of unseen-domain AUROC on label-shift level and leakage, with recipe as a covariate (variance inflation ≈ 1.04), gives label-shift coefficients of -0.166 $[-0.217, -0.115]$ (Fitzpatrick17k), -0.107 $[-0.148, -0.065]$ (MED-NODE) and -0.172 $[-0.221, -0.123]$ (SD-198) for a change from no shift to full shift at fixed leakage [PENDING: confirm shift coding and resampling unit]. On `pad_heldout` the coefficient is $+0.041$ $[-0.033, +0.110]$. Comparing runs at equal leakage across recipes (A4 at shift 100%, leakage 0.78, against the main recipe at shift 0%, leakage 0.79) gives the same direction (+0.23 to +0.26).

A natural example without DANN. Label shift alone can invert a class-conditional Mahalanobis score. Training on BCN and treating HAM as OOD, Mahalanobis AUROC is already 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, while k-NN gives 0.658. HAM-heldout is 67% nevus against 31% in BCN test; reweighting HAM to BCN's class mix gives 0.489 $[0.453, 0.524]$, and no class-matched AUROC differs from 0.5 (post hoc). In the main setting the OOD sets are *less* nevus-heavy than ISIC, so detection works at $\lambda_{\text{adv}} = 0$; DANN creates the condition by moving them toward nevus.

4.3. Mechanism: misaligned classes behind chance-level leakage

Unfamiliar images move to the majority class. At $\lambda_{\text{adv}} = 2$ with full label shift, the share of `pad_heldout` images whose nearest class centroid is nevus rises from 0.28 to 0.63, and to 0.75–0.85 on the unseen sets; without label shift it stays at 0.12 and 0.39–0.49.

Images of non-nevus classes are pulled into the nevus region: split by true class, non-nevus OOD images fall to AUROC 0.42 (main recipe) and 0.47 (A4), whereas images still nearest a non-nevus centroid stay at 0.62 and 0.64. Because pooled AUROC is the class-share-weighted mean of per-class AUROCs, and nevus is half the ISIC test set, this displacement is enough to push the pooled score below chance.

The displacement follows the domain direction. For each OOD image we measured how far it lies along the direction separating ISIC from PAD in the $\lambda_{\text{adv}} = 0$ model, and how much closer to its nearest ID centroid it moves at $\lambda_{\text{adv}} = 2$. The two correlate on every lesion dataset (Spearman 0.33–0.47), while the deviation orthogonal to that direction does not predict the pull ($|\rho| \leq 0.07$ on four of five sets). Between two $\lambda_{\text{adv}} = 0$ models of different seeds the same correlation is smaller (0.11–0.29); the DANN-specific excess exceeds 0.15 on all four unseen datasets (0.18–0.33) but not on `pad_heldout` (0.057), and is larger in the pre-projection backbone (0.24–0.45 on all five).

Chance-level leakage hides the misalignment. The domain probe of Table 1 is scored partly on images the adversary trained on. On held-out images a linear probe reads the domain at chance from $\lambda_{\text{adv}} = 0.5$ on (0.502–0.506), but a k-NN probe still reads it (0.55–0.61), and under label shift an unweighted probe misreports invariance (Table 4). With full label shift DANN reaches 0.510 on an unweighted held-out probe, yet within each class the domain remains decodable (0.618), more so than in any other condition. Correcting the class mix the adversary aligns (shift 0%, or importance weighting with the true priors) raises the unweighted reading (0.63–0.64) while lowering the within-class one (0.56–0.57). Under label shift, DANN matches the marginal distributions by misaligning classes, as the theory of Zhao et al. (2019) implies; leakage should therefore be measured within class (post hoc).

4.4. Clinical consequence and mitigation

Melanomas called benign. With a threshold accepting 95% of ISIC validation images, almost all unseen-domain malignant images are accepted at every λ_{adv}, so the monitor does not catch them in either case. What changes is the classifier: the share of melanomas predicted nevus with confidence above 0.9 rises from 7.5% to 48.7% on Fitzpatrick17k and from 8.1% to 81.2% on SD-198 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ [PENDING: n and CIs per cell; label map].

No fix at scoring time. Reversing the sign of the score is not a remedy: at $\lambda_{\text{adv}} \geq 0.5$ it reaches only 0.55–0.63 on `pad_heldout` and Fitzpatrick17k with a false-positive rate of 0.84–0.95 at 95% true-positive rate [PENDING: verify FPR range in s_flip], and at $\lambda_{\text{adv}} = 0$ it would turn 0.808 into 0.192. Relative Mahalanobis (Ren et al., 2021) and per-class standardised distances stay below 0.5 on all three unseen datasets at $\lambda_{\text{adv}} = 2$.

Importance-weighted alignment. IWDAN reweights the source in the adversarial loss by estimated target class priors, and how well it works depends on the estimate. *BBSE* priors are far from the truth (total variation 0.65–0.76), plausibly because its assumption of unchanged class-conditional inputs fails across these domains; IWDAN then keeps leakage at the DANN level (0.642 vs 0.634) and recovers 32% of the lost AUROC on `pad_heldout`

Table 4: Domain leakage on held-out images under three probes, and unseen-domain detection ($\lambda_{\text{adv}} = 2$, main recipe, last checkpoint). Unweighted: as usually reported; class-balanced: both domains reweighted to a common class mix; within-class: class-size-weighted mean of per-class probes. Fitz: Mahalanobis AUROC on Fitzpatrick17k.

condition	unweighted	class-balanced	within-class	Fitz
DANN, shift 100%	0.510	0.602	0.618	0.340
DANN, shift 0%	0.629	0.539	0.560	0.691
IWDAN (BBSE)	0.534	0.562	0.597	0.508
IWDAN (50 labels)	[PENDING:]	[PENDING:]	0.583	0.741
IWDAN (true priors)	0.635	0.537	0.576	0.717

(0.513) and 45% on Fitzpatrick17k (0.508), while MED-NODE (0.350) and SD-198 (0.488) stay at or below chance. *EM* estimation was unstable: in two of five seeds it placed mass on classes absent from PAD, so we treat it as not having run correctly. With priors from 50 *labelled target images* (total variation 0.10–0.14), detection on the unseen domains returns to the level obtained with the true priors (Fitzpatrick17k 0.741 vs 0.717, SD-198 0.734 vs 0.693, MED-NODE 0.606 vs 0.563; `pad_heldout` 0.619 vs 0.647), and within-class leakage is lower than under DANN (0.58 vs 0.62). Neither the 50-label nor the true-prior variant meets our condition for a fix, because unweighted leakage rises (0.81 and 0.78 against 0.63), and both cost in-distribution balanced accuracy (0.39–0.40 against 0.48). Across estimators, the error in the estimated priors did not predict detection as a rank correlation (Spearman -0.33, $[-0.80, +0.28]$), with the EM runs contributing invalid priors [PENDING: IWCDAN, CDAN, $\lambda_{\text{adv}} = 0.5$ cells; per-method details to appendix].

5. Discussion

An inversion, not a degradation. The central result is not that out-of-distribution detection weakens under adversarial invariance training but that the ordering reverses. Degradation would be unsurprising — a representation stripped of domain information should be worse at recognising a domain, and chance is what that argument predicts, which is what a single-encoder model reaches at $\lambda_{\text{adv}} = 0.25$ (0.525). Going *past* chance is not predicted by it, and is what Section 4.3 accounts for. The results then compose into one failure sequence: a model trained with invariance to acquisition domain, as recommended, is deployed at a new site, where it increasingly predicts the benign majority class — actinic keratosis recall $0.245 \rightarrow 0.016$ — with higher confidence than on its own test data, while the monitor that should flag those images ranks them as more routine than familiar ones and every metric the developer watches moves favourably.

The failure belongs to gradient reversal, not to invariance. Reducing leakage is not sufficient to produce it, and in this sweep leakage is not even achieved: it bottoms out at 0.555 against a floor of 0.5 and rises again at the largest weights while detection stays inverted. MMD reduces leakage along the same axis and collapses nothing at the levels it reaches with the classifier intact, where gradient reversal has already inverted. The two also

fail in opposite manners: pushed harder, MMD becomes less accurate *and* less confident and its out-of-domain calibration error falls, while under gradient reversal accuracy falls as confidence rises and calibration error triples. One degrades honestly, the other is confidently wrong. Wald et al. (Wald et al., 2021) showed that enforcing multi-domain calibration *produces* invariance; the converse fails here, which suggests that where invariance is the goal, calibration-based or moment-matching objectives may be preferable because they leave the out-of-domain uncertainty signal intact. We offer that as a hypothesis: MMD never reached the invariance level gradient reversal reaches without destroying the task, and we tested one adversarial formulation, so whether CDAN or adversarial discriminative adaptation behaves the same way is not established here.

The danger zone is where the method succeeds. Risk scales with *achieved* invariance, not with the nominal weight or the presence of an adversarial term. Where the adversary never learned, leakage stayed at 0.94 across four orders of magnitude of λ_{adv} and the safety failure did not appear either. A weak adversary produces neither debiasing nor harm — so many deployed configurations may be harmless, and a configuration's safety cannot be read off its hyperparameters.

Tolerance without generalisation. Full-spectrum OOD detection (Yang et al., 2023) prescribes that covariate-shifted in-distribution inputs be accepted rather than flagged. Our models achieve that prescription and are unsafe because of it: they accept out-of-domain images more readily than their own test data while classifying them at 0.254 balanced accuracy against a floor of 0.167. Acceptance and competence are separate properties, and current protocols evaluate them separately or not at all. Any claim of tolerance to covariate shift should carry cross-domain accuracy on the same inputs, since tolerating inputs a model cannot classify is a failure mode rather than a success.

Recommendation. Report OOD detection and out-of-domain calibration **as a function of achieved invariance** rather than at a single operating point; report cross-domain accuracy alongside any claim of tolerance to covariate shift; and state, when a separate dataset serves as the OOD set, that this measures covariate shift rather than semantic novelty. Practitioners tuning λ_{adv} already train at several values, so this changes what is reported rather than what is run. It converts an invisible failure into a visible number.

Monitor outside the adversarial path. Since the collapse is a property of the representation that received adversarial gradient, one mitigation is to score novelty on a representation that did not. A frozen ImageNet ResNet-50 gives Mahalanobis AUROC 0.988 on Fitzpatrick17k and 0.833 on `pad_heldout`, against the trained model's 0.764 and 0.843 at $\lambda_{\text{adv}} = 0$ and 0.391 and 0.427 at $\lambda_{\text{adv}} = 2$. It never inverts, but **we had pre-registered** 0.90 **in all four cells and** `pad_heldout` **does not reach it**, where the frozen encoder merely matches the untrained model rather than improving on it. A partial mitigation, not a fix; retreating to an earlier layer of the *same* encoder does not work at all (Section 4.3).

The inversion is deepened, not created, in the minority classes. Per-class AUROC, which unlike the pooled value does not depend on the ID class mix, is 0.14–0.39 on `pad_heldout` and 0.10–0.34 on Fitzpatrick17k at $\lambda_{\text{adv}} = 2$ for every non-nevus class. Two are already at or below chance beforehand: on `pad_heldout` at $\lambda_{\text{adv}} = 0$, actinic keratosis is

0.499 and squamous cell carcinoma 0.444. Gradient reversal deepens an inversion that already exists for the rarest and most clinically consequential classes, and the pooled AUROC at $\lambda_{\text{adv}} = 0$ conceals this because nevus is half the in-distribution set.

The mechanism, and what remains open. Gradient reversal removes the acquisition signal by moving out-of-domain images into the region the majority class occupies (Section 4.3) — an efficient way to satisfy the adversary, since the densest part of the latent space is where domain information is hardest to recover, and simultaneously the worst outcome for a distance-based novelty score, which reads density as familiarity. The two objectives are not merely in tension: invariance is obtained by the move that defeats detection. What remains open is why gradient reversal takes this route and moment matching does not, why the synthetic reproduction failed, and whether the majority class is special or merely the densest available — a dataset with two comparably sized dominant classes would separate the last of these.

6. Limitations

Absolute cross-domain performance is poor for every model here. Plain accuracy on `pad_heldout` is 0.245 for the factorized model and 0.391 for the baseline against a 0.385 majority rate, so no model here is deployable. Our claim concerns the *effect of* λ_{adv}, and the deployment pathway above is a mechanism of harm rather than a description of a fielded system.

The inversion is observed only across imaging modality. ISIC is dermoscopy while PAD-UFES and Fitzpatrick17k are clinical photography, so the main contrast confounds site with modality and Fitzpatrick17k does not resolve it. The same-modality contrast within ISIC (Section 4.2) separates the claims: the collapse reproduces between two hospitals, the inversion does not. We cannot say whether that reflects the size of the shift, modality gaps specifically, or these datasets.

Three attempts to replicate outside dermatology returned no result, for the same underlying reason in different guises: the manipulation did not take, so there was nothing to measure. On three-hospital Camelyon17 the adversary head never trained; on iWildCam it trained but the encoder never became invariant; and on Camelyon17 restricted to *two* hospitals — the obvious diagnosis, matching the dermatology domain count, with the gradient path verified first — the adversary reached chance ($\ln 2$) while a post-hoc probe still read the hospital at 0.92–0.95, a fall of at most 0.037 against the 0.10 we had pre-committed. The pre-registered stopping rule returns a negative and we stop. **Domain count is not the barrier**, and we cannot say what is; the pattern — an adversary driven to chance while the information it was asked to remove stays linearly decodable — is a known failure mode of adversarial removal (Elazar and Goldberg, 2018). **We report no inversion outside dermatology**; a single-seed sub-chance reading there did not replicate (0.534 ± 0.050 over three seeds). What those runs do support is the dissociation in Section 4.3. Details in Appendix A.4.

Generality beyond dermatology remains untested. **Backbone coverage.** Comparable invariance was not reached with a ResNet-50 backbone, so whether the inversion extends beyond EfficientNet-B3 is untested. Three of the nine runs failed their adversary gate outright and are reported as null experiments rather than negative results. **The objective**

comparison is bounded. MMD never reached the invariance level gradient reversal attains; it could only approach it by destroying the classifier. We therefore cannot say what moment-matching would do at matched invariance, only that at the leakage levels it reaches, gradient reversal has already inverted and MMD has not.

Detector non-independence. The five scoring rules are functions of one 16-dimensional latent; the backbone analysis and the single-encoder replication carry the independent evidence. **Calibration arithmetic.** Out-of-domain ECE is close to mean confidence minus accuracy here, so those three results are one phenomenon viewed three ways. **Seeds.** Five at every λ_{adv} on the leak-free split, and three at two interior points of the image-level sweep in the appendix; effect sizes are large relative to seed variance, but we claim no formal test across seven points.

Splits. The main sweep uses the image-level ISIC split, in which 3,041 of 5,067 test images (60.0%) share a `lesion_id` with training or validation. This inflates in-distribution balanced accuracy by 0.20 ($0.496 \rightarrow 0.692$ at $\lambda_{\text{adv}} = 0$), so the absolute values in Table 1 are optimistic by that amount; Section 4.1 reports the lesion-disjoint replication at five seeds, under which every conclusion is unchanged. PAD splits are patient-level, disjoint by patient and by lesion. **Third-domain coverage.** 3,887 of 16,577 Fitzpatrick17k images were obtained; the set is web-scraped under mixed copyright, used for evaluation only, and not redistributed.

7. Conclusion

Under domain-adversarial training, out-of-distribution detection does not merely weaken. It reverses: a model asked which inputs are unfamiliar answers with its own test data and ranks a new clinic's images as the more routine of the two. The reversal appears at the smallest adversarial weight that achieves invariance, holds on a dataset the adversary never saw, reproduces without the factorized architecture, and does not appear under moment-matching objectives that reduce leakage along the same axis.

It is invisible to everything a developer monitors. In-distribution accuracy holds, in-distribution calibration is unchanged and the domain leakage probe improves exactly as intended — under an image-level and a lesion-disjoint ISIC split alike — while confidence on unseen domains overtakes confidence on familiar data and recall for a premalignant class falls to 0.016.

The cause is identifiable: invariance is obtained by moving out-of-domain images into the majority class's region of the latent space, which is exactly the move that makes a density-based novelty score call them familiar. Acting on it does not require the explanation. Reporting out-of-distribution detection, out-of-domain calibration and cross-domain accuracy as functions of achieved invariance costs almost nothing and makes the failure visible.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

References

Alceu Bissoto, Eduardo Valle, and Sandra Avila. Debiasing skin lesion datasets and models? Not so fast. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020.

Konstantinos Bousmalis, George Trigeorgis, Nathan Silberman, Dilip Krishnan, and Dumitru Erhan. Domain separation networks. In *Advances in Neural Information Processing Systems*, 2016.

Roxana Daneshjou, Kailas Vodrahalli, Roberto A. Novoa, Melissa Jenkins, Weixin Liang, Veronica Rotemberg, Justin Ko, Susan M. Swetter, Elizabeth E. Bailey, Olivier Gevaert, Pritam Mukherjee, Michelle Phung, Kiana Yekrang, Bradley Fong, Rachna Sahasrabudhe, Johan A. C. Allerup, Utako Okata-Karigane, James Zou, and Albert S. Chiou. Disparities in dermatology AI performance on a diverse, curated clinical image set. *Science Advances*, 8(32), 2022.

Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Conference on Empirical Methods in Natural Language Processing*, pages 11–21, 2018.

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.

Matthew Groh, Caleb Harris, Luis Soenksen, Felix Lau, Rachel Han, Aerin Kim, Arash Koochek, and Omar Badri. Evaluating deep neural networks trained on clinical images in dermatology with the Fitzpatrick 17k dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2021.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems*, 2018.

Zachary C. Lipton, Yu-Xiang Wang, and Alexander J. Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.

Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems*, 2020.

Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems*, 2018.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019.

Andre G. C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, Fábio C. R. Alves Jr, José G. M. Esgario, Alana C. Simora, Pedro B. C. Castro, Felipe B. Rodrigues, Patricia H. L. Frasson, Renato A. Krohling, Helder Knidel, Maria C. S. Santos, Rachel B. do Espírito Santo, Telma L. S. G. Macedo, Tania R. P. Canuto, and Luiz F. S. de Barros. PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32, 2020.

Jie Ren, Stanislav Fort, Jeremiah Liu, Abhijit Guha Roy, Shreyas Padhy, and Balaji Lakshminarayanan. A simple fix to Mahalanobis distance for improving near-OOD detection. *arXiv preprint arXiv:2106.09022*, 2021.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1): 21–41, 2002.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning*, 2022.

Remi Tachet des Combes, Han Zhao, Yu-Xiang Wang, and Geoffrey Gordon. Domain adaptation with conditional distribution matching and generalized label shift. In *Advances in Neural Information Processing Systems*, 2020.

Duc Vinh Tran. Your OOD monitor has seen these patients: Group overlap miscalibrates deployment monitors in medical imaging, 2027. Under review at MIDL 2027. Co-author list not final.

Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5, 2018. See also Combalia et al., BCN20000, arXiv:1908.02288, 2019, and the ISIC 2019 challenge.

Yoav Wald, Amir Feder, Daniel Greenfeld, and Uri Shalit. On calibration and out-of-domain generalization. In *Advances in Neural Information Processing Systems*, 2021.

Ximei Wang, Mingsheng Long, Jianmin Wang, and Michael I. Jordan. Transferable calibration with lower bias and variance in domain adaptation. In *Advances in Neural Information Processing Systems*, 2020.

Julia K. Winkler, Christine Fink, Ferdinand Toberer, Alexander Enk, Teresa Deinlein, Rainer Hofmann-Wellenhof, Luc Thomas, Aimilios Lallas, Andreas Blum, Wilhelm Stolz,

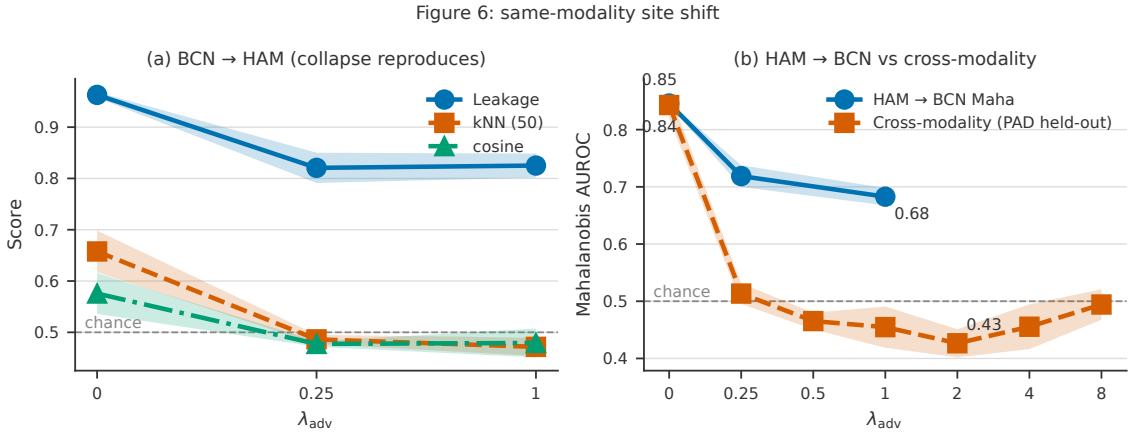


Figure 2: **A same-modality site shift separates the two claims.** (a) Barcelona to HAM10000: leakage falls and the detectors with headroom cross chance. (b) With the roles reversed to restore headroom, Mahalanobis falls from a baseline matching the cross-modality sweep but stops above chance, while the cross-modality sweep continues below it.

and Holger A. Haenssle. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. *JAMA Dermatology*, 155(10):1135–1141, 2019.

Xinheng Wu, Jie Lu, Zhen Fang, and Guangquan Zhang. Meta OOD learning for continuously adaptive OOD detection. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. doi: 10.1109/ICCV51070.2023.01773.

Jingkang Yang, Kaiyang Zhou, and Ziwei Liu. Full-spectrum out-of-distribution detection. *International Journal of Computer Vision*, 2023.

Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Kaiyang Zhou, Wayne Zhang, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*, 2023.

Han Zhao, Remi Tachet des Combes, Kun Zhang, and Geoffrey Gordon. On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *PMLR*, pages 7523–7532, 2019.

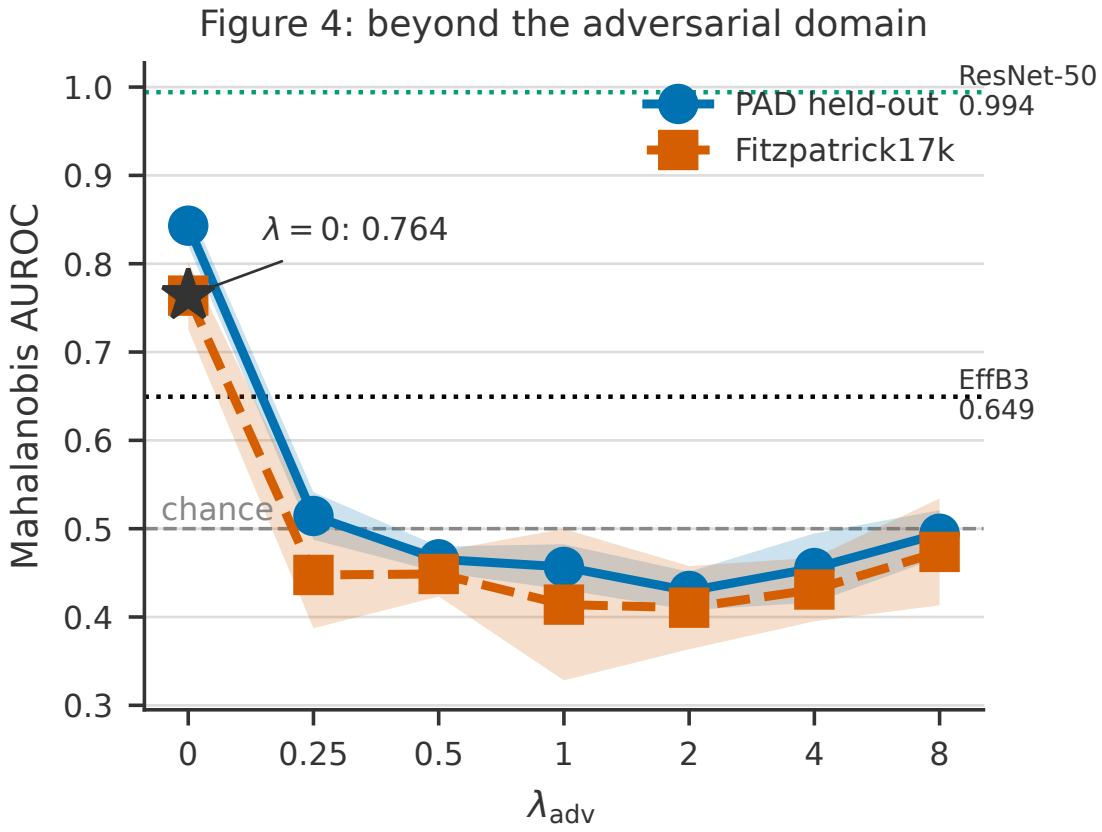


Figure 3: **The collapse is not confined to the adversarial domain.** AUROC against λ_{adv} for the held-out PAD partition and for Fitzpatrick17k, which is absent from every training branch. Horizontal references give the ResNet-50 baseline and the EfficientNet-B3 control on Fitzpatrick; the controlled comparison is the $\lambda_{\text{adv}} = 0$ point of the sweep itself.

Appendix A. Supporting tables and figures

A.1. The image-level split

Table 8 is the same sweep under the image-level ISIC split used by most published work on this dataset, in which 60.0% of test images share a `lesion_id` with training. Every pattern is the same; the in-distribution accuracies are about 0.20 higher throughout.

A.2. Confidence, accuracy and calibration on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted (0.930 versus 0.910).

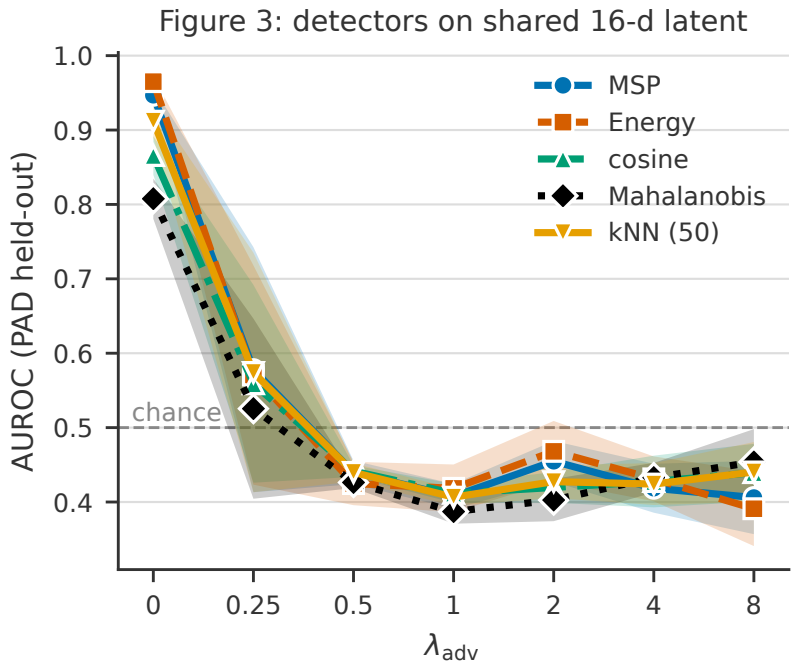


Figure 4: **Every scoring rule moves together.** The five detectors against λ_{adv} on the held-out PAD partition, mean ± 1 s.d. All five are functions of the same 16-dimensional latent and are therefore not independent evidence; Figure 6 carries the replication on a different representation.

Cross-domain diagnostic accuracy does not improve at any λ_{adv}: balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{adv} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls 0.245 → 0.184, beneath the 0.385 majority-class rate throughout. Macro AUC is roughly preserved (0.605 → 0.581), so discrimination survives while the decision rule does not.

Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and actinic keratosis recall falls from 0.245 to 0.016 (49/202 → 3/202 images, mean over five seeds). Squamous cell carcinoma moves the same way (0.088 → 0.021, 6/66 → 1/66) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

In this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy, which is why the confidence, accuracy and calibration results are not independent corroboration of each other.

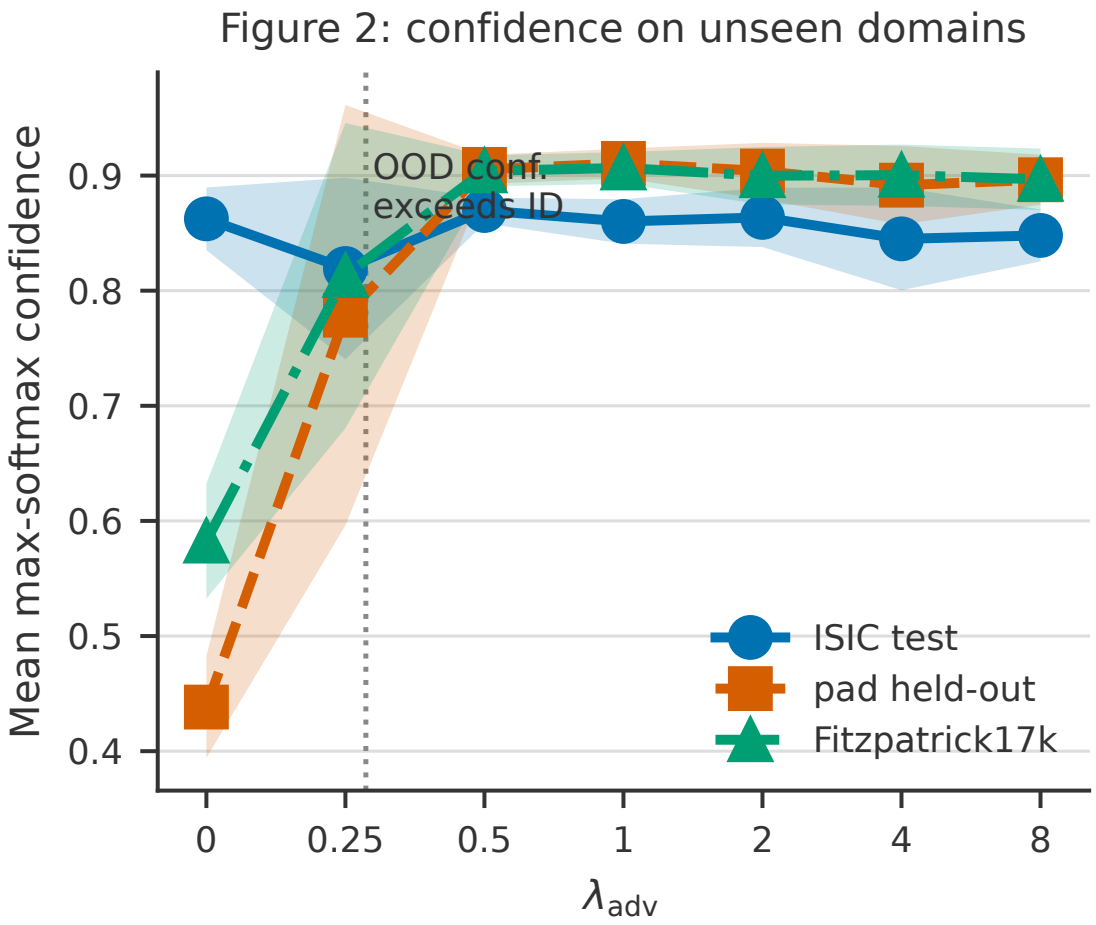


Figure 5: **Confidence on unseen domains overtakes confidence on familiar data.** Mean maximum-softmax probability against λ_{adv} for the ISIC test split, the held-out PAD partition and Fitzpatrick17k. The crossing occurs by $\lambda_{\text{adv}} \approx 0.5$. λ spacing is categorical.

A.3. Single-encoder replication

Plain single-encoder domain-adversarial training on the same data and splits. Every detector collapses at $\lambda_{\text{adv}} = 0.25$ — k-NN 0.944 $\rightarrow$ 0.528, energy 0.911 $\rightarrow$ 0.507, cosine 0.884 $\rightarrow$ 0.509, MSP 0.868 $\rightarrow$ 0.515 — and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014. In-distribution accuracy is flat throughout (0.845 $\rightarrow$ 0.832) and cross-domain accuracy falls from 0.390 to a plateau near 0.32.

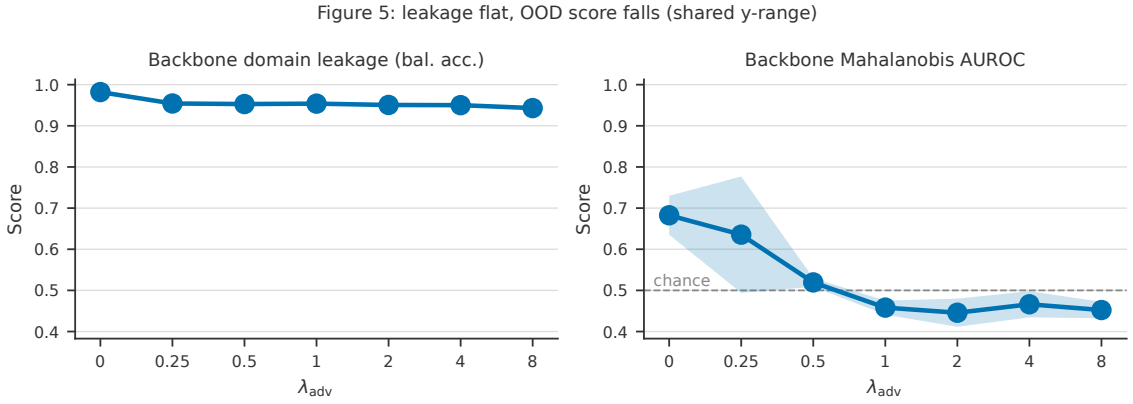


Figure 6: **Leakage does not mediate the collapse.** At the pre-projection backbone the domain probe stays flat across the sweep while Mahalanobis AUROC falls. Domain information remains linearly decodable and detection degrades regardless.

Table 5: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
No access to PAD labels					
ERM[§]	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{adv} = 0.25$	**0.763**	**0.525**	**0.845**[†]	—
Uses PAD labels — not comparable on cross-domain accuracy					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 6.
[§] scored on `pad_full` with a uniform optimiser and an ISIC-only loss; the λ_{adv}=0 row of Table 6 uses `pad_heldout`, grouped learning rates and `ignore_index`. They are not the same condition. [PENDING: harmonise or relabel.]

A.4. External replication attempts

Camelyon17, three hospitals. The adversary head never trained: cross-entropy sat at $\ln 3$ from the first epoch at every λ_{adv} while a post-hoc probe on the same features read 0.94, despite a gradient graph verified correct (head cosine +1.000, encoder −1.000).

Table 6: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k-NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	**0.944**	0.868	0.884	**0.911**	0.845	0.390
0.25	3	**0.763**	0.525	**0.528**	0.515	0.509	**0.507**	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

Table 7: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	`pad_heldout` 0.427 vs `pad_adv` 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.432 vs 0.427 on `pad_heldout`; reweighting *strengthens* the inversion to 0.259
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{\text{adv}}$	0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, same model and data
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	4.879 $\rightarrow$ 4.500 while leakage 0.914 $\rightarrow$ 0.553 (same three runs); decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already 4.5$\times$ tighter at $\lambda_{\text{adv}} = 0$; the adversary *decompresses* it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion (λ_{adv}=0: 0.693; λ_{adv}=2: 0.786)

iWildCam. The adversary did train — best-epoch cross-entropy 3.13 against $\ln 243 = 5.49$ — but the encoder never became invariant: the binary train-versus-OOD site probe moved only from 0.911 to 0.891 across $\lambda_{\text{adv}} \in [0, 10]$ against a floor of 0.5, and no detector collapsed.

Camelyon17, two hospitals. Pre-committed in `PRECOMMIT_CAMELYON2.json` before running. The adversary reached chance (final cross-entropy 0.692 against $\ln 2 = 0.693$, from 0.11 at $\lambda_{\text{adv}} = 0$) and in-distribution accuracy held at 0.995, but the hospital probe fell only $0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$, inside its own seed spread, so the leakage gate failed and the verdict is negative. The exploratory seeds added afterwards do not reopen it: Mahalanobis on the held-out hospital reads 0.534 ± 0.050 at $\lambda_{\text{adv}} = 1$ over seeds 42–44, above chance, so the single-seed 0.478 did not replicate. These runs appear in Section 4.3

Table 8: The intervention under the image-level ISIC split, with the two non-adversarial controls. Five seeds at $\lambda_{\text{adv}} \in \{0, 0.25, 1, 2, 8\}$, three at $\lambda_{\text{adv}} \in \{0.5, 4\}$.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.571 ± 0.027	0.692	0.100	0.694	0.509 ± 0.023	0.460 ± 0.057	0.295 ± 0.028
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	5	0.591 ± 0.035	0.695	0.097	0.735	0.457 ± 0.026	0.414 ± 0.086	0.258 ± 0.012
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391

for the non-mediation claim only, where the five detectors move 0.12–0.22 while the probe moves 0.007.

A.5. Same-modality site shift: probe validation and the HAM direction

The sites are separable, and not for trivial reasons. A logistic probe on frozen ERM features distinguishes BCN from HAM at balanced accuracy 0.962 ± 0.007 against a floor of 0.5. Masking the dermatoscope vignette reduces this only to 0.943 ± 0.005, so the border is not the signal. Hand-crafted colour and intensity statistics reach 0.901 at best — close to the 0.876 ceiling the same features reach on ISIC↔PAD, so this pair is no more low-level than the original contrast — while sharpness statistics sit at chance (0.532). A probe given only the ground-truth class label reaches 0.668, reflecting the two sites' different class distributions; the feature probe is far above that reference.

Mahalanobis is uninformative in the BCN-in-distribution direction. It reads 0.379 at $\lambda_{\text{adv}} = 0$, already below chance before any adversarial training. Reweighting the in-distribution set to the out-of-domain class mix raises it to 0.487, so most of that baseline inversion is class composition — HAM is two-thirds nevus and therefore concentrated near a single well-populated centroid — with a residual we attribute to HAM10000's more standardised acquisition and correspondingly lower feature variance. This is why the reversed direction, which restores headroom, carries the scope claim in Section 4.2. In that direction the class-reweighted variant falls from 0.74 to 0.54, and Mahalanobis reads 0.719 ± 0.019 at $\lambda_{\text{adv}} = 0.25$.

A.6. A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a monitor is not in itself difficult. What distinguishes the dual-encoder's context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999, two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.